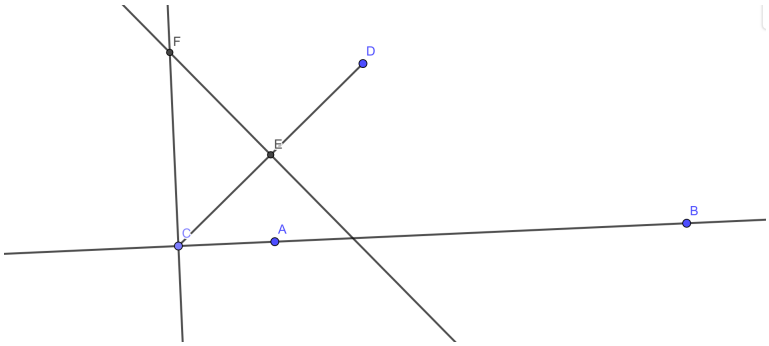
Math 2001
Geogebra

Tracing some Familiar Curves

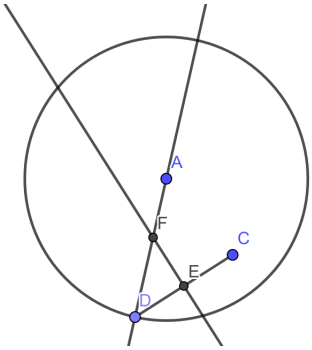
1. Use Geogebra to recreate the construction below with D as $(0, 4)$ and $\overleftrightarrow{AB}$ as $y = 2$.



- $\overleftrightarrow{AB}$
- C as a free point on $\overleftrightarrow{AB}$
- D not on $\overleftrightarrow{AB}$ and $\overline{DC}$
- E as the midpoint of $\overline{DC}$
- A line through E perpendicular to $\overline{DC}$
- A line through C perpendicular to $\overleftrightarrow{AB}$
- F as the intersection of the two perpendicular lines

Animation: Highlight F and from the settings menu choose to show trace for F. Highlight C and press the play button in the algebra pane to animate the point. What type of function you have traced, can you find its equation?

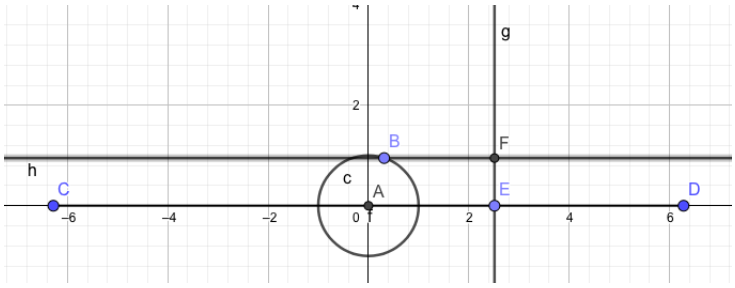
2. Use Geogebra to recreate the construction below with A as $(0, 0)$ and C as $(2, 0)$. For your construction let the circle centered at A have radius 3.



- Circle centered at A
- C interior to the circle distinct from A
- D as a free point on the circle
- $\overleftrightarrow{AD}$ and $\overline{CD}$
- E as the midpoint of $\overline{CD}$
- The perpendicular bisector of $\overline{CD}$
- F as the intersection of the perpendicular bisector of $\overline{CD}$ and $\overleftrightarrow{AD}$

Animation: Highlight F and from the settings menu choose to show trace for F. Highlight D and press the play button in the algebra pane to animate the point around the circle. What type of curve have you traced, can you find its equation?

3. Create the picture below in Geogebra with the properties that follow in the Graphics perspective. In the settings menu you can change the x-axis units in terms of π.



- The circle is centered at the origin with radius 1, have Geogebra plot this exactly by entering the appropriate equation in the Algebra pane.
- B is a free point on the circle.
- C is $(-2\pi, 0)$ and D is $(2\pi, 0)$, again have Geogebra plot these exactly.
- E is a free point on $\overline{CD}$
- Line g is perpendicular to $\overline{CD}$ through E
- Line h is perpendicular to g through B
- F is the intersection of g and h

Animation: Drag point E to the origin and then point B to $(1, 0)$. In Settings-Algebra for E change the speed to .5 and in settings for F check *Show Trace*. For the next entry in the Algebra pane enter *StartAnimation*(B,E). A play/pause button will appear on the bottom left of your sketch. Press play, what function have you sketched and why?